

Assignmet 5

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● acalaria@ArushiSharma:~$ echo $HADOOP_HOME
/home/acalaria/PYPrograms/hadoop-3.4.2
● acalaria@ArushiSharma:~$ echo $HADOOP_CONF_DIR
/home/acalaria/PYPrograms/hadoop-3.4.2/etc/hadoop
● acalaria@ArushiSharma:~$ which hadoop
/home/acalaria/PYPrograms/hadoop-3.4.2/bin/hadoop
● acalaria@ArushiSharma:~$ hadoop version
Hadoop 3.4.2
Source code repository https://github.com/apache/hadoop.git -r 84e8b89ee2ebe6923691205b9e171badde7a495c
Compiled by ahmarsu on 2025-08-20T10:30Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum fa94c67d4b4be021b9e9515c9b0f7b6
This command was run using /home/acalaria/PYPrograms/hadoop-3.4.2/share/hadoop/common/hadoop-common-3.4.2.jar
● acalaria@ArushiSharma:~$ nano ~/.bashrc
● acalaria@ArushiSharma:~$ nano ~/PYPrograms/hadoop-3.4.2/etc/hadoop/core-site.xml
● acalaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/core-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>fs.defaultFS</name>
        <value>hdfs://localhost:9000</value>
    </property>

</configuration>
● acalaria@ArushiSharma:~$ nano ~/PYPrograms/hadoop-3.4.2/etc/hadoop/hdfs-site.xml
● acalaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/hdfs-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>dfs.replication</name>
        <value>1</value>
    </property>
```

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● acalaria@ArushiSharma:~$ nano ~/PYPrograms/hadoop-3.4.2/etc/hadoop/hdfs-site.xml
● acalaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/hdfs-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>dfs.replication</name>
        <value>1</value>
    </property>

    <property>
        <name>dfs.namenode.name.dir</name>
        <value>file:///home/acalaria/hadoop_data/namenode</value>
    </property>

    <property>
        <name>dfs.datanode.data.dir</name>
        <value>file:///home/acalaria/hadoop_data/datanode</value>
    </property>

</configuration>
● acalaria@ArushiSharma:~$ ls ~/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml*
/home/acalaria/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml
● acalaria@ArushiSharma:~$ cp ~/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml.template \
~/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml
cp: cannot stat '/home/acalaria/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml.template': No such file or directory
● acalaria@ArushiSharma:~$ nano ~/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml
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● acelaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>mapreduce.framework.name</name>
        <value>yarn</value>
    </property>

</configuration>
● acelaria@ArushiSharma:~$ nano ~/PYPrograms/hadoop-3.4.2/etc/hadoop/yarn-site.xml
● acelaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
    </property>

</configuration>
● acelaria@ArushiSharma:~$ cat ~/PYPrograms/hadoop-3.4.2/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

    <property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
    </property>

</configuration>

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● acelaria@ArushiSharma:~$ mkdir -p ~/hadoop_data/namenode
mkdir -p ~/hadoop_data/datanode
● acelaria@ArushiSharma:~$ ls -l ~/hadoop_data
total 8
drwxr-xr-x 2 acelaria acelaria 4096 Sep 17 14:17 datanode
drwxr-xr-x 2 acelaria acelaria 4096 Sep 17 14:17 namenode

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• acalaria@ArushiSharma:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ArushiSharma]
ArushiSharma: Warning: Permanently added 'arushisharma' (ED25519) to the list of known hosts.
2026-09-17 14:19:30,721 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
• acalaria@ArushiSharma:~$ jps
30600 DataNode
30874 SecondaryNameNode
30444 NameNode
31055 Jps
• acalaria@ArushiSharma:~$ hdfs dfs -mkdir -p /user/$USER/hadoop_exp5
2026-09-17 14:20:01,039 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
• acalaria@ArushiSharma:~$ hdfs dfs -ls /user/$USER
2026-09-17 14:20:07,067 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - acalaria supergroup 0 2026-09-17 14:20 /user/acalaria/hadoop_exp5
• acalaria@ArushiSharma:~$ echo -e "Hadoop HDFS Experiment 5\nNameNode manages metadata\nDataNode stores data blocks\nHDFS provides distributed storage" > ~/sample.txt
• acalaria@ArushiSharma:~$ cat ~/sample.txt
Hadoop HDFS Experiment 5
NameNode manages metadata
DataNode stores data blocks
HDFS provides distributed storage
• acalaria@ArushiSharma:~$ hdfs dfs -put ~/sample.txt /user/$USER/hadoop_exp5/
2026-09-17 14:21:07,416 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
• acalaria@ArushiSharma:~$ hdfs dfs -ls /user/$USER/hadoop_exp5
2026-09-17 14:21:12,826 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 acalaria supergroup 113 2026-09-17 14:21 /user/acalaria/hadoop_exp5/sample.txt

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• acalaria@ArushiSharma:~$ hdfs dfs -cat /user/$USER/hadoop_exp5/sample.txt
2026-09-17 14:21:36,988 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Hadoop HDFS Experiment 5
NameNode manages metadata
DataNode stores data blocks
HDFS provides distributed storage
• acalaria@ArushiSharma:~$ hdfs dfs -cp /user/$USER/hadoop_exp5/sample.txt /user/$USER/hadoop_exp5/sample_copy.txt
2026-09-17 14:22:01,318 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
• acalaria@ArushiSharma:~$ hdfs dfs -ls /user/$USER/hadoop_exp5
2026-09-17 14:22:06,923 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 acalaria supergroup 113 2026-09-17 14:21 /user/acalaria/hadoop_exp5/sample.txt
-rw-r--r-- 1 acalaria supergroup 113 2026-09-17 14:22 /user/acalaria/hadoop_exp5/sample_copy.txt
• acalaria@ArushiSharma:~$ hdfs dfs -get /user/$USER/hadoop_exp5/sample_copy.txt ~/sample_download.txt
2026-09-17 14:22:28,316 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
• acalaria@ArushiSharma:~$ cat ~/sample_download.txt
Hadoop HDFS Experiment 5
NameNode manages metadata
DataNode stores data blocks
HDFS provides distributed storage
• acalaria@ArushiSharma:~$ hdfs dfs -df -h
2026-09-17 14:22:51,690 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Filesystem Size Used Available Use%
hdfs://localhost:9000 1006.9 G 24.2 K 905.4 G 0%
• acalaria@ArushiSharma:~$ hdfs dfs -rm /user/$USER/hadoop_exp5/sample_copy.txt
2026-09-17 14:23:23,098 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /user/acalaria/hadoop_exp5/sample_copy.txt
• acalaria@ArushiSharma:~$ hdfs dfs -ls /user/$USER/hadoop_exp5
2026-09-17 14:23:27,428 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 acalaria supergroup 113 2026-09-17 14:21 /user/acalaria/hadoop_exp5/sample.txt

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acelaria@ArushiSharma:~$ wecho "First additional Hadoop file" > ~/file1.txt
echo "Second additional Hadoop file" > ~/file2.txt

hdfs dfs -mkdir -p /user/$USER/hadoop_exp5_multiple

hdfs dfs -put ~/file1.txt ~/file2.txt /user/$USER/hadoop_exp5_multiple/

hdfs dfs -ls /user/$USER/hadoop_exp5_multiple
Command 'wecho' not found, did you mean:
  command 'echo' from deb coreutils-from-gnu (0.0.0-ubuntu25)
  command 'echo' from deb coreutils-from-uutils (0.0.0-ubuntu25)
  command 'echo' from deb coreutils-from-busybox (0.0.0-ubuntu25)
  command 'echo' from deb coreutils-from-toybox (0.0.0-ubuntu25)
  command 'aecho' from deb atalkd (4.2.3-ds-2.1ubuntu0.2)
Try: sudo apt install <deb name>
2026-09-17 14:35:34,688 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-17 14:35:35,960 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-17 14:35:37,619 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 acelaria supergroup      0 2026-09-17 14:35 /user/acelaria/hadoop_exp5_multiple/file1.txt
-rw-r--r-- 1 acelaria supergroup    30 2026-09-17 14:35 /user/acelaria/hadoop_exp5_multiple/file2.txt
acelaria@ArushiSharma:~$

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acelaria@ArushiSharma:~$ hdfs dfs -mkdir -p /user/$USER/hadoop_exp5_copy

hdfs dfs -cp /user/$USER/hadoop_exp5_multiple/file1.txt /user/$USER/hadoop_exp5_copy/

hdfs dfs -ls /user/$USER/hadoop_exp5_copy
2026-09-17 14:36:05,965 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-17 14:36:06,974 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-17 14:36:08,111 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 acelaria supergroup      0 2026-09-17 14:36 /user/acelaria/hadoop_exp5_copy/file1.txt
acelaria@ArushiSharma:~$

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acelaria@ArushiSharma:~$ hdfs dfs -rm -r /user/$USER/hadoop_exp5_copy

hdfs dfs -ls /user/$USER
2026-09-17 14:36:27,748 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /user/acelaria/hadoop_exp5_copy
2026-09-17 14:36:28,826 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
drwxr-xr-x - acelaria supergroup      0 2026-09-17 14:23 /user/acelaria/hadoop_exp5
drwxr-xr-x - acelaria supergroup      0 2026-09-17 14:35 /user/acelaria/hadoop_exp5_multiple
acelaria@ArushiSharma:~$

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